Experiment 12

Z Test, T Test, F Test and ANOVA.

Program:

T-test

```
import numpy as np
from scipy import stats

# Sample data
marks = np.array([72, 68, 75, 70, 74, 69, 71, 73, 70, 72])

# Hypothesized mean
mu_0 = 70

# One-sample t-test
t_stat, p_value = stats.ttest_1samp(marks, mu_0)

print(f"T-statistic: {t_stat:.3f}")
print(f"P-value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
    print("Reject Null Hypothesis → Mean is significantly different from 70.")
else:
    print("Fail to Reject Null Hypothesis → No significant difference.")
```

Z-test

```
import numpy as np
from math import sqrt
from scipy.stats import norm

# Given data
x_bar = 51.2 # sample mean
mu_0 = 50 # population mean
sigma = 3 # population standard deviation
n = 36 # sample size

# Calculate Z-statistic
z_stat = (x_bar - mu_0) / (sigma / sqrt(n))

# Two-tailed p-value
p_value = 2 * (1 - norm.cdf(abs(z_stat)))

print(f"Z-statistic: {z_stat:.3f}")
print(f"P-value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
    print("Reject Null Hypothesis → Mean is significantly different from 50 g.")
else:
    print("Fail to Reject Null Hypothesis → No significant difference.")
```

Anova Test

```
import numpy as np
from scipy import stats

# Data
A = [20, 22, 23]
B = [19, 20, 18]
C = [25, 27, 26]

# Perform one-way ANOVA
f_stat, p_value = stats.f_oneway(A, B, C)

print(f"F-statistic: {f_stat:.3f}")
print(f"P-value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
    print("Reject Null Hypothesis → Means are significantly different.")
else:
    print("Fail to Reject Null Hypothesis → No significant difference.")
```

Output

```
T-test Output:
T-statistic: 1.993
P-value: 0.0774
Fail to Reject Null Hypothesis → No significant difference.

Z-test Output:
Z-statistic: 2.400
P-value: 0.0164
Reject Null Hypothesis → Mean is significantly different from 50 g.

ANOVA / F-test Output:
F-statistic: 25.923
P-value: 0.0011
Reject Null Hypothesis → Means are significantly different.
```